

Talos

Spoken instructions for learned dinner-table tasks

AI Infra Summit Hackathon
September 2026



Two SO-101 arms

Six learned skills

Bottle relays in both directions

OpenVINO inference

Speechmatics voice input

A physical simulation prototype.

Broader seven-item control remains unfinished.

Six skills in one continuous scene

"Set the table"



- 01 **Place the bottle**
- 02 Place the plate
- 03 Place the mug
- 04 Open the drawer
- 05 Place the fork
- 06 Place the spoon

The human voice rehearsal completed all six skills with 15 live mug corrections.

Learned motion with separate physical checks

The network generates movement

Camera features condition neural joint trajectories.
Measured joint feedback regulates task progress.

The selected profile adds stereo observation and a learned local correction model during late mug placement.

Speechmatics

Transcribes supported spoken commands.

Task grammar and RGB checks

Choose the next supported skill.

MuJoCo physical monitor

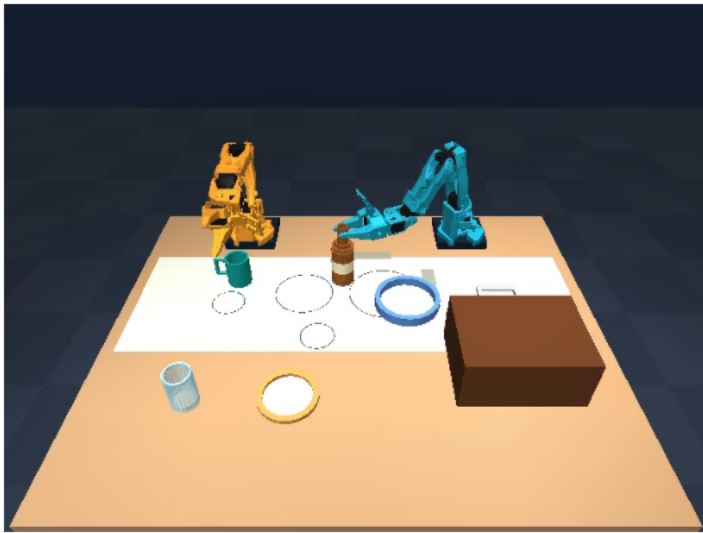
Checks contact, placement and parked arms. It does not generate learned actions.

201 verified inputs for five newer dinner skills

52 verified inputs for four relay models

A bottle relay through the shared table

One arm releases the bottle before the other takes over.



Blue arm lifts

12.0 simulated seconds

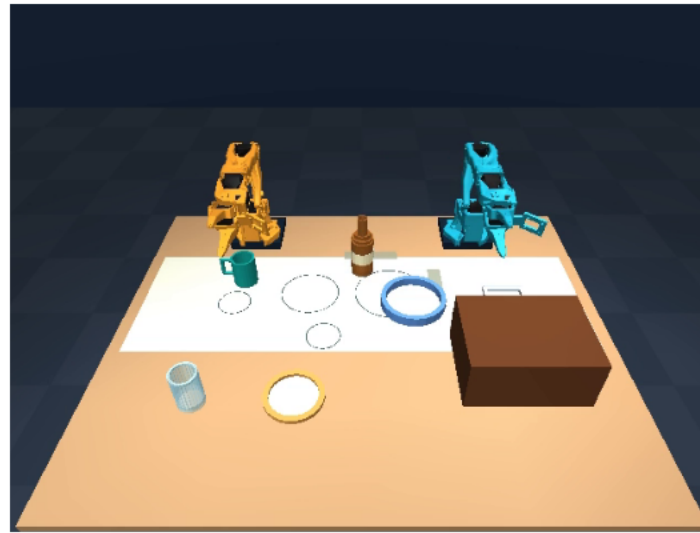
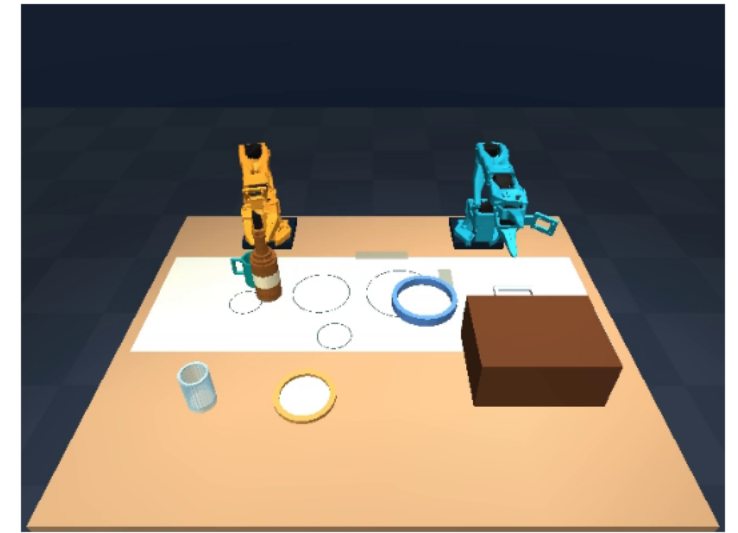


Table release and parking

37.0 simulated seconds



Amber arm carries

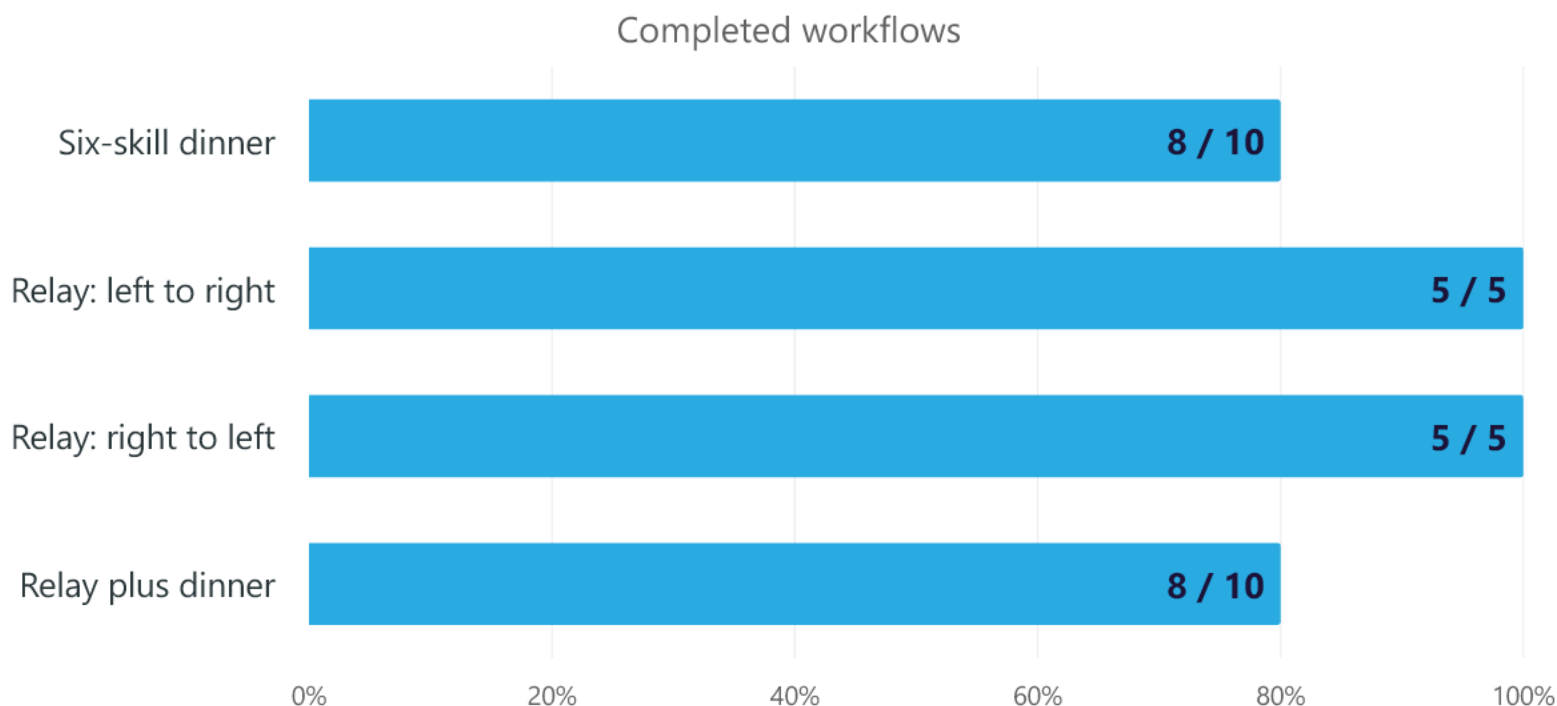
60.0 simulated seconds

5 / 5 left to right 5 / 5 right to left

Upright start regions, fixed goals.
Transfer uses table contact.

Frozen trials retain both successes and failures

Limited starting presets and fixed destinations define these results.



Two original dinner failures

14.59 mm

Mug placement XY error

13.61 mm

Spoon placement XY error

The dinner placement threshold stays at 8 mm. Each chart row has its own frozen protocol.

Visual correction during late mug placement

The same correction model receives live or frozen images.



Updated images matter for the correction model

The original baseline also passes every final case. This does not establish a higher success rate than that baseline.

All 96 development and final workflows remain preserved.

Two narrow presets. Broad visual recovery remains unfinished.

Intel execution on the available laptop

Six learned tasks complete with Intel CPU or iGPU inference and Intel UHD rendering.

Core i7-10850H

Legacy Intel hardware
Tested on seed 42
Recorded physical outcomes

Historical original-bottle network timing

0.157 ms

Intel CPU, FP32 inference

0.688 ms

Intel UHD iGPU, FP32 inference

Network-only medians.
These are not full robot
control-loop timings.

Core Ultra Series 2/3 eligibility remains unresolved.
The latest live-mug profile also needs its separate Intel verification run.

Broader seven-item control remains unfinished

**Jointly randomized objects,
full tabletop rotation and
stable alternative orientations**

Calibrated RGB-D and adapted CLIPort run on the RTX 4070. The shared Talos action policy is untrained.

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Complete seven-item
workflows passed

Next capability milestone

Reliable physical grasps and routes, then varied whole-scene demonstrations and a shared camera-driven policy.

github.com/jannissio/talos-ai-infra

Code, models, compact inputs and all outcomes. MIT for original work.

Private CPU demo verified.
Public access awaits final release.